

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of M. Pharm (DRA)

University Theory Examination December 2016

PHRA001 PHARMACEUTICAL FORMULATION DEVELOPMENT

Date: 16/12/2016, Friday

Time: 10:00 a. m. to 01:00 p. m.

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1** What are particulate carriers? Describe methods for preparation of nanoparticles. **10**
- Q.2** Discuss in detail the physicochemical and biological factors influencing design of controlled release systems. **10**

OR

- Q.2** What is a stability indicating assay method? Discuss the important aspects involved in development of stability indicating assay method? **10**
- Q.3 Answer the following (Any three)** **15**
- (A) Write note on solid dispersion.
- (B) Define and classify liposomes? Describe their limitations.
- (C) Write note on osmotic controlled drug delivery system.
- (D) Explain the need for improvement in drug solubilization.

SECTION-II

- Q.4** Define preformulation. Write objectives of preformulation studies. Discuss in detail preformulation studies to be carried out for Oral solid dosage form. **10**
- Q.5** Bioavailability of a poorly soluble drug is challenging to preformulation scientist. Discuss physical and chemical modification of API and use of excipients to overcome the issue. **10**

OR

- Q.5 (A)** Define intrinsic dissolution. Discuss comparison of dissolution profile by model dependent methods. **05**
- (B)** Discuss various factors that influence dissolution. **05**
- Q.6 Answer the following (Any three)** **15**
- (A) Discuss various methods in preparation of Fast dissolving Tablets
- (B) Justify DSC and FTIR are important analytical tools in preformulation studies
- (C) Write short note on BCS classification and its significance
- (D) Discuss Self Emulsifying Drug Delivery System

